

TARA: Detecting and Classifying Exoplanet Transits in Noisy TESS Light Curves via a Dual CNN–RandomForest Pipeline

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Transit Analysis and Recognition AI

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ABSTRACT—Transiting-exoplanet discovery from space photometry is limited less by classification skill than by the difficulty of separating shallow, noisy transit signals from instrumental and astrophysical false positives. We present **TARA** (Transit Analysis and Recognition AI), an end-to-end system that ingests a raw TESS or Kepler light curve, searches it for periodic transits, and classifies the candidate with two complementary models: a five-seed one-dimensional convolutional neural network (CNN) that reads the phase-folded flux directly, and a 400-tree RandomForest (RF) operating on twelve physically motivated features. On held-out data the CNN reaches a planet-vs-not ROC AUC of 0.951 (0.902 ± 0.017 under honest five-fold cross-validation) and the RF reaches a planet-vs-rest AUC of 0.953 across four classes (transit, eclipsing binary, blend, noise). A blind test on 331 real TESS Objects of Interest recovers 50% of known planets end-to-end; through ablation we show that the dominant loss is the single-sector *detection* front-end, not the classifier, and that the models are *data-bound* rather than architecture-bound. TARA is served through a lightweight FastAPI backend with explicit honesty guards that abstain rather than fabricate a verdict on low-quality data.

INDEX TERMS—exoplanets, transit photometry, TESS, convolutional neural networks, random forest, light-curve classification, model calibration.

I. INTRODUCTION

The *Kepler* and *TESS* missions have transformed exoplanet science by measuring the brightness of hundreds of thousands of stars, watching for the periodic dimming that occurs when a planet crosses (“transits”) its host star [1], [2]. A transit is a shallow event: a Jupiter produces roughly a 1% dip, an Earth-analogue closer to 0.01%. At that depth the signal competes with instrumental systematics, stellar variability, and—most dangerously—*astrophysical false positives* such as eclipsing binaries and background blends that mimic a planetary dip. TESS compounds the challenge: it tiles almost the whole sky but observes most stars for only a single ~ 27 -day sector, leaving few transits per target.

Manual vetting cannot scale to the millions of candidate signals produced. Deep learning has therefore become central: AstroNet [3] classified *Kepler* threshold-crossing events (TCEs), later work extended the approach to TESS [8], and ExoMiner [7] set a modern benchmark. A common thread is that these systems classify *pre-detected* TCEs supplied by a separate pipeline—they do not perform the search themselves.

The hard part is not spotting a dip but explaining it. A periodic dimming can be a genuine planet, but equally a grazing or diluted eclipsing binary, a fainter background eclipsing system whose light is blended into the target aperture, or nothing astrophysical at all—an instrumental artefact or a beat of stellar variability. These classes overlap in depth and period, and the features that separate them—secondary eclipses, odd-even depth differences,

centroid motion during the dip—are subtle. A system that answers only “planet or not” discards this structure; one that names the impostor gives a vetter something to act on. TARA therefore adopts a four-way taxonomy—transit, eclipsing binary, blend, noise—mirroring how human vetters reason, and pairs it with a search front-end so the whole judgement runs from a raw light curve.

The problem addressed here is AI-based detection of exoplanets from noisy light curves. TARA answers it end-to-end: from a raw light curve it performs *both* the transit search and the classification, and reports a calibrated, honesty-guarded verdict. Our contributions are: (i) a dual CNN + RandomForest classifier combining a data-driven and a physics-driven view; (ii) a four-class taxonomy that explicitly models blends, not just planet-vs-not; (iii) “honesty guards” that abstain on unreliable input; and (iv) an evidence-based analysis showing the deployed system is *detection-limited* and *data-bound*, which directs where future effort should go.

II. PROBLEM DEFINITION

Given a star identifier (or an uploaded light curve), TARA solves two coupled sub-tasks. **Detection:** does the star show a periodic, transit-like dimming above the noise? **Classification:** if so, is the signal a planetary *transit*, an *eclipsing binary* (EB), a *blend* (a diluted or background eclipse), or *noise*? The output is a class label, a calibrated confidence, and—when the evidence is weak—an explicit *uncertain* / *abstain* state rather than a forced guess.

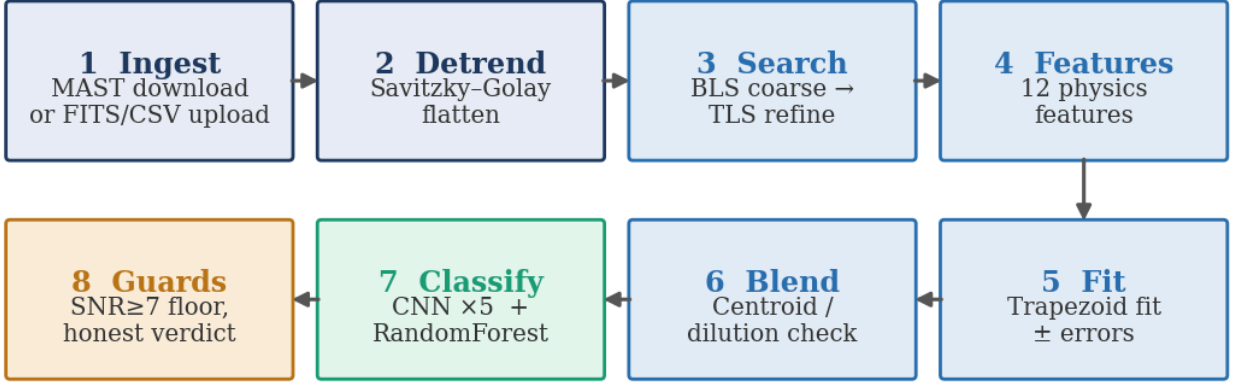


Fig. 1. The TARA eight-stage pipeline. A raw light curve is ingested, detrended, searched for a period (a coarse Box Least Squares scan refined by Transit Least Squares), reduced to twelve features and a trapezoid fit, checked for blends, then classified by both models before the honesty guards produce the final verdict.

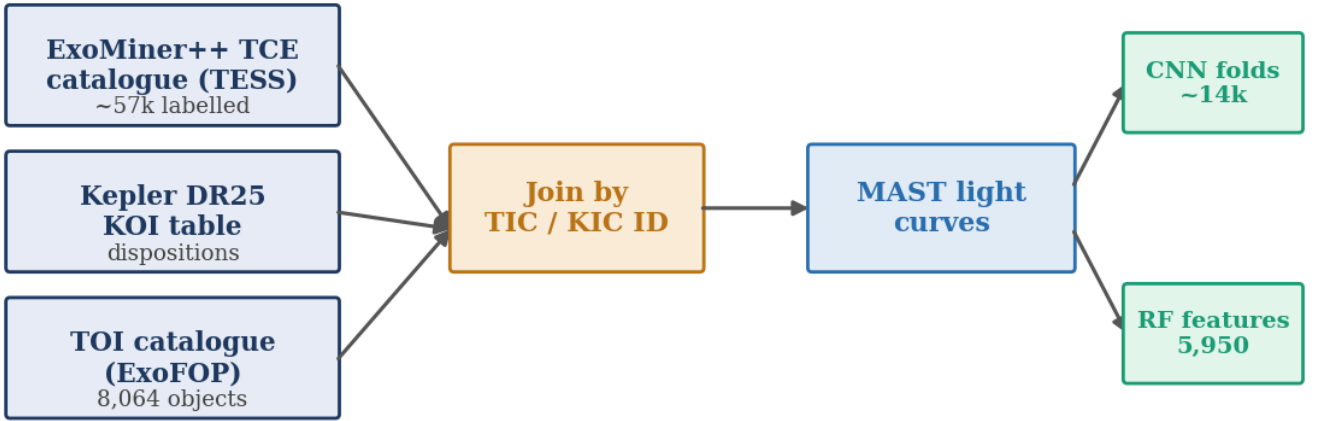
III. DATA AND PROVENANCE

Every training example needs three things joined together (Fig. 2): a *label* (what the object is), an *ephemeris* (period and epoch, so the curve can be phase-folded), and a *light curve*. Labels and ephemerides come from vetted catalogues; the flux comes from the Mikulski Archive for Space Telescopes (MAST), pulled with `lightcurve` [6] and joined to the labels by TIC (TESS) or KIC (Kepler) identifier.

We draw labels from three catalogues: the ExoMiner-derived labelled TCE catalogue (TESS, ~57k rows), the *Kepler* DR25 KOI dispositions, and the ExoFOP TOI catalogue (8,064 objects). We

deliberately do *not* train on the TIC or Candidate Target List star catalogues: they contain neither dispositions nor light curves and are therefore unusable as supervision. Light curves are used at single-sector cadence, preferring SPOC/TESS-SPOC products where available.

The labelled examples feed two training sets: roughly 14,000 phase-folded views for the CNN and 5,950 twelve-feature rows for the RF. One data decision proved pivotal. *Kepler*'s 30-minute long-cadence smears transit ingress and egress; including those transits in the positive class taught the model a softened transit shape that transferred poorly to TESS's sharp 2-minute cadence. Excluding them ("variant B") is analysed in Section VI-C.



Labels come from vetted catalogues; flux comes from MAST — joined by star ID.

Fig. 2. Dataset provenance. Vetted catalogues supply labels and ephemerides; MAST supplies flux. The two are joined by star ID and split into CNN folds and RF feature rows. Star catalogues (TIC/CTL) are excluded because they carry no labels or curves.

IV. SYSTEM ARCHITECTURE

TARA is a compact scientific pipeline (Fig. 1) exposed through a FastAPI service and a vanilla HTML/JavaScript dashboard. The stages are: (1) **Ingest** a light curve from MAST or an uploaded FITS/CSV; (2) **Detrend** with a Savitzky-Golay filter to remove slow stellar and instrumental trends; (3) **Search** for a period using a coarse Box Least Squares [5] scan on ten-minute-binned

flux, refined by Transit Least Squares (TLS) [4] within $\pm 3\%$; (4) **Features**—twelve physically motivated quantities; (5) **Fit** a trapezoid transit model with bootstrap uncertainties; (6) **Blend** checks via centroid motion and dilution; (7) **Classify** with the CNN ensemble and the RF; and (8) **Guards**, which enforce a detection floor and emit an honest verdict. The whole backend is ~1,000 lines of Python; the intelligence lives in the two trained models.

V. MODELS AND TRAINING

A. Convolutional neural network (detection view)

The CNN is a one-dimensional network that reads the amplitude-normalised phase-folded flux directly—the raw “shape” of the candidate—and outputs a single probability that the signal is planetary (a binary head). To reduce variance we train five independent seeds on a mixed *Kepler*+TESS corpus and average their logits. Because ensembles are systematically over-confident, we apply temperature scaling [10] with $T = 1.55$, which recalibrates the probabilities without altering their ranking (Section VI-B).

B. RandomForest (physics view)

The RF (400 trees) classifies all four categories from twelve interpretable features grouped as: transit *shape* (depth, duration, duration fraction, V-shape), signal *significance* (SNR, number of transits, period), false-positive *discriminators* (odd–even depth difference, secondary-eclipse ratio, centroid offset), and *noise* descriptors (out-of-transit scatter, point-to-point RMS). Figure 5 shows the model’s own Gini importances: transit depth (0.117), duration fraction (0.112), SNR (0.108) and period (0.107) dominate—reassuringly, the features a human vetter would weigh most. The model is evaluated with a *grouped* hold-out so that no star appears in both training and test, preventing leakage.

C. Honesty guards

Two models are not enough if the input is junk. TARA enforces an $\text{SNR} \geq 7$ detection floor (a field-standard threshold), a zero-centred-flux guard that rejects corrupt products, and a “confident” flag that requires the top class probability ≥ 0.40 and a margin ≥ 0.10 over the runner-up. When these are not met, the interface reports *uncertain* and the CNN reading is suppressed rather than shown as a spuriously confident number.

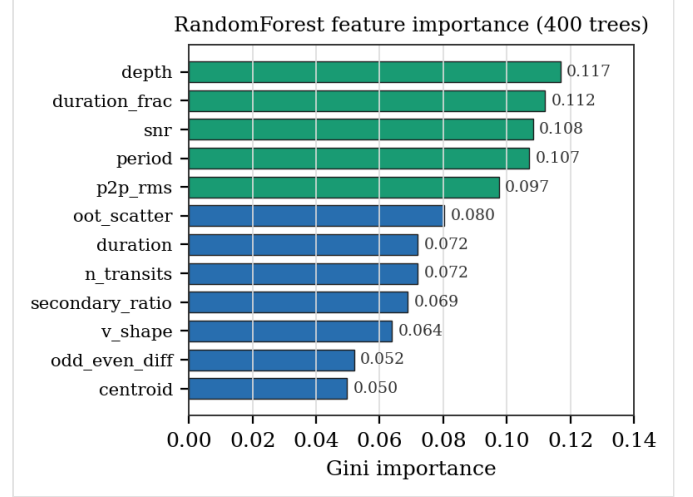


Fig. 5. RandomForest Gini feature importances, read directly from the trained model (400 trees, 12 features). Physically meaningful transit-shape and significance features lead; centroid and odd–even differences contribute the fine false-positive discrimination.

VI. EXPERIMENTS AND RESULTS

A. CNN discrimination

On held-out folds the CNN reaches a planet-vs-not ROC AUC of **0.951** combined—0.932 on *Kepler* and 0.966 on TESS (Fig. 3a). We report a more conservative 0.902 ± 0.017 under five-fold cross-validation as the honest floor. The higher TESS score is the operationally relevant one, since TESS is the target survey.

B. Calibration

Temperature scaling lowers the expected calibration error from 0.038 to 0.025 (Fig. 3b) while leaving AUC unchanged at 0.949—by construction, since a monotonic temperature transform cannot reorder predictions. Calibration makes the confidence numbers trustworthy, which the honesty guards then rely on.

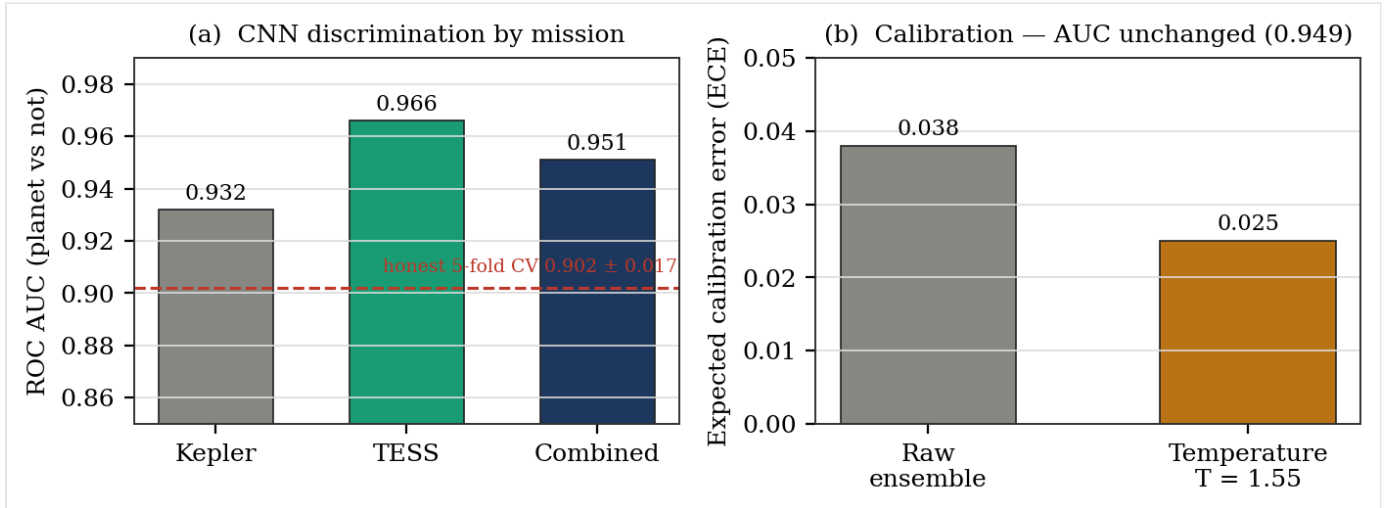


Fig. 3. (a) CNN ROC AUC by mission, with the honest five-fold cross-validation floor (dashed). (b) Temperature scaling reduces calibration error (ECE) without changing AUC.

C. The transit-bias ablation

Our central data finding is shown in Fig. 4. Training the RF with *Kepler* long-cadence transits in the positive class (variant A) yields a planet-vs-rest AUC of 0.906; removing them (variant B) raises it to **0.953**, a gain of +0.047. A concrete regression illus-

trates why: for the real non-planet TIC 149623590, variant A assigned transit probability 0.68 (a false positive), while variant B assigns 0.33 and correctly favours the eclipsing-binary class. Variant B is deployed.

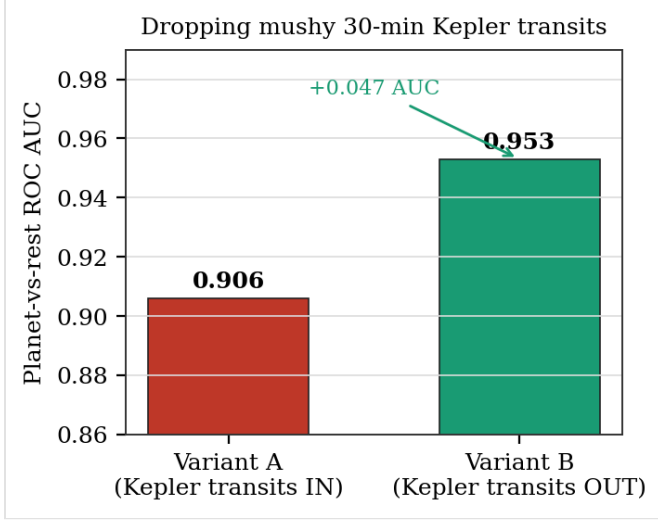


Fig. 4. Ablation: excluding mushy 30-minute *Kepler* transits from the transit class removes a shape bias and improves planet-vs-rest AUC from 0.906 to 0.953.

D. RandomForest four-class performance

On the grouped hold-out the RF reaches 75.8% overall accuracy and a 0.953 planet-vs-rest AUC (Fig. 6). Transit precision/recall are 0.76/0.69. The hardest class is *blend* (recall 0.67): diluted eclipses are intrinsically close to shallow transits, and labelled blends are scarce—a data problem we return to in Section VII.

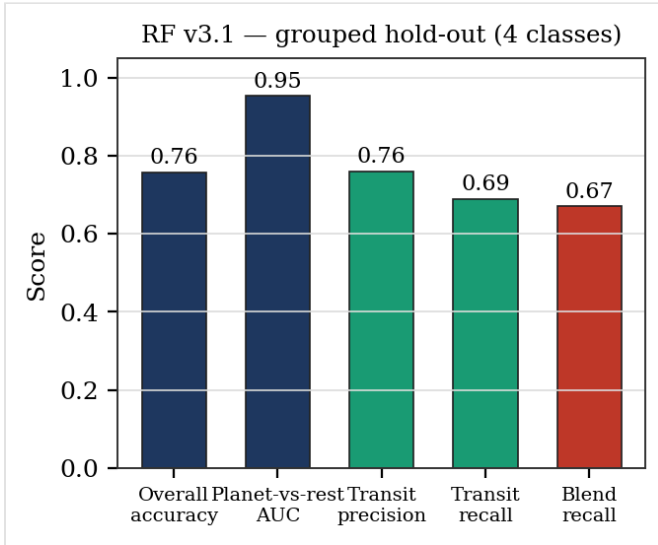


Fig. 6. RF v3.1 headline metrics on the grouped hold-out. Planet detection (transit) is strong; blend recall is the weakest link.

E. Hyper-parameter tuning is exhausted

To test whether more tuning would help, we ran a controlled sweep. Under five-fold cross-validation (0.902 ± 0.017) every variation—heavier regularisation, a wider network, longer training—landed within the ± 0.017 noise band; disabling data augmentation actively hurt (0.889); and the five-seed ensemble was the *only* change that produced a real, repeatable gain. The conclusion is that TARA is **data-bound, not architecture-bound**: further accuracy will come from more and cleaner labels, not from tuning knobs.

F. Blind end-to-end test

We ran the full pipeline blind on 331 real TESS Objects of Interest (Fig. 7). Of 200 known planets, 50% were recovered end-

to-end as *transit*; of 80 eclipsing binaries, 22% were labelled EB; and 74% of the 50 false-positives / other were correctly rejected. The planet number looks modest until it is decomposed—which is the subject of the next section.

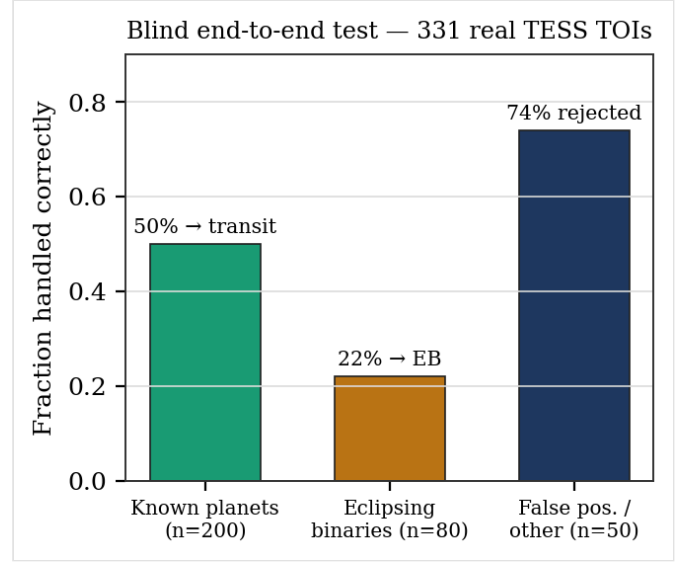


Fig. 7. Blind end-to-end recovery on 331 real TESS TOIs, by ground-truth disposition. False-positive rejection is strong; EB recall reflects domain shift from the mostly-*Kepler* EB training data.

VII. DISCUSSION

A. Detection, not classification, is the bottleneck

Figure 8 decomposes the 50% planet number. Of 200 known planets, 128 (64%) pass the single-sector detection floor, and 101 (50%) are ultimately called *transit*. Critically, 72 planets (36%) are lost at the *detection* stage—their single-sector signal never clears $\text{SNR} \geq 7$ —while only 27 are lost at classification. The end-to-end 50% therefore *understates* the classifier: given a detected signal, TARA labels it correctly the large majority of the time.

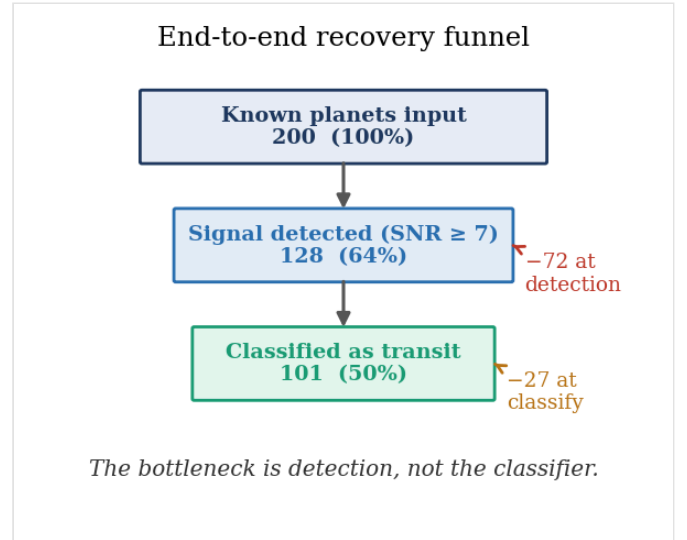


Fig. 8. Where known planets are lost. The dominant drop is at detection (single-sector SNR floor), not at the classifier—evidence that the front-end search, not the model, caps end-to-end recovery.

B. Fair comparison to ExoMiner

ExoMiner reports ROC AUC ≈ 0.95 , but on *pre-detected, multi-sector* TCEs—it never searches. TARA's *classification* AUC

(0.951 CNN, 0.953 RF) is the same task and is competitive. TARA's 50% is a strictly harder, single-sector, *end-to-end* figure that includes the search ExoMiner is handed for free. Compared like-for-like on classification, the systems are in the same regime.

C. Honesty as a design goal

Because scientific users must trust an abstention as much as a detection, TARA is built to say “uncertain” rather than fabricate a verdict on zero-signal or corrupt data. The guards in Section V-C are part of the architecture, not an afterthought.

D. Limitations

The system is single-sector by default (the detection cap above); EB recall on real TESS is depressed by training mostly on *Kepler* EBs (domain shift); the blend class is data-starved; and TLS search is CPU-heavy, motivating an offline batch mode.

VIII. CONCLUSION AND FUTURE WORK

TARA is an exoplanet detector and four-class classifier that reaches ~ 0.95 classification AUC and recovers half of known planets blind from single-sector TESS data, with the remaining loss traced to detection rather than the model. The evidence points to clear next steps: (i) add a “classify pre-detected TCEs” mode that bypasses the search bottleneck (expected end-to-end recovery 75–85%); (ii) inject the TESS Eclipsing Binary catalogue [11] and the *Kepler* Certified-False-Positive flags to strengthen the EB and blend classes; (iii) add multi-sector stitching for a batch pipeline; and (iv) add a difference-imaging / pixel branch—the true modality gap to ExoMiner. As curated survey data becomes available, the highest-value action is simply to re-train on it, exactly as the data-bound analysis predicts.

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